Oracle SQL Developer Web / Database Actions – Tools and Authentication Scenarios Overview

Oracle Database Actions, formerly known as SQL Developer Web, is a web-based interface running on Oracle REST Data Services and provides development, data tools, administration, and monitoring features for Oracle Database on-premises and Oracle Database Cloud services. Here all the authentication scenarios, prerequisites and the Database Actions services in the Cloud are presented.

Oracle Database Actions: Autonomous AI Database

Prerequisites to access Database Actions:

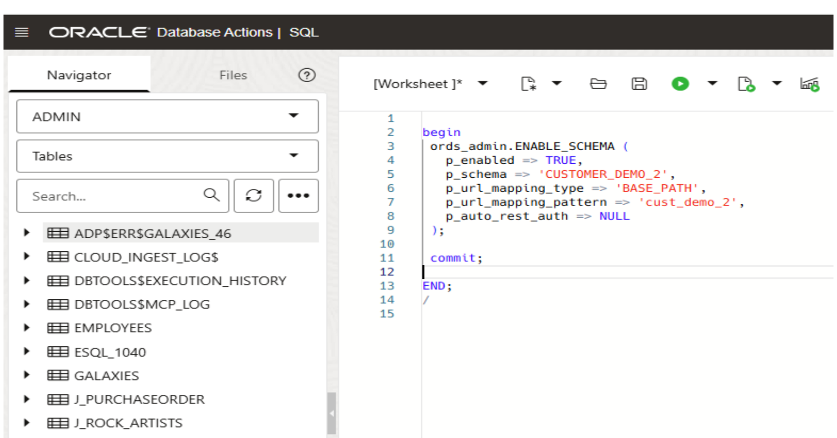
In Oracle Autonomous Database, Database Actions is already built-in and the administrator user ADMIN is already enabled to authenticate. To enable a given schema user to authenticate, it has to be REST-enabled as follows, connected by ADMIN:

- From User Administration detail page:

```
-- REST ENABLE
BEGIN
     ORDS_ADMIN.ENABLE_SCHEMA(
         p_enabled => TRUE,
         p_schema => 'CUSTOMER_DEMO_2',
         p_url_mapping_type => 'BASE_PATH',
         p_url_mapping_pattern => 'customer_demo_2',
         p_auto_rest_auth=> FALSE
     );
     -- ENABLE DATA SHARING
     C##ADP$SERVICE.DBMS_SHARE.ENABLE_SCHEMA(
             SCHEMA_NAME => 'CUSTOMER_DEMO_2',
             ENABLED => TRUE
     );
     commit;
END;
/
```

Note: parameter value p_schema is in upper case (“**CUSTOMER_DEMO_2**”).

- From SQL Worksheet via PL/SQL code:



NB: enabling user via PL/SQL in SQL Worksheet allows you to choose a *schema_alias* name (***p_url_mapping_partner*** API attribute) different from the schema name.

In order to authenticate to Database Actions, other than being REST-enabled, to have at least minimal operational capabilities the schema user also needs the followings:

- Non-zero value on Tablespace Quota (unless it is a read-only user);
- CONNECT, RESOURCE assigned roles

For example, connected as ADMIN user and checking the roles assigned to a schema user (again, for uer *CUSTOMER_DEMO_2*):

```
select * from dba_role_privs where grantee = 'CUSTOMER_DEMO_2'
order by granted_role;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULT SQL HISTORY TASK MONITOR

All rows fetched: 4 in 0.087 seconds

	GRANTEE	GRANTED_ROLE	ADMIN_OPTION	DELEGATE_OPTION	DEFAULT_ROLE	COMMON	INHERITED
1	CUSTOMER_DEMO_2	CONNECT	NO	NO	YES	NO	NO
2	CUSTOMER_DEMO_2	DB_DEVELOPER_ROLE	NO	NO	NO	NO	NO
3	CUSTOMER_DEMO_2	DWROLE	NO	NO	NO	NO	NO
4	CUSTOMER_DEMO_2	RESOURCE	NO	NO	YES	NO	NO

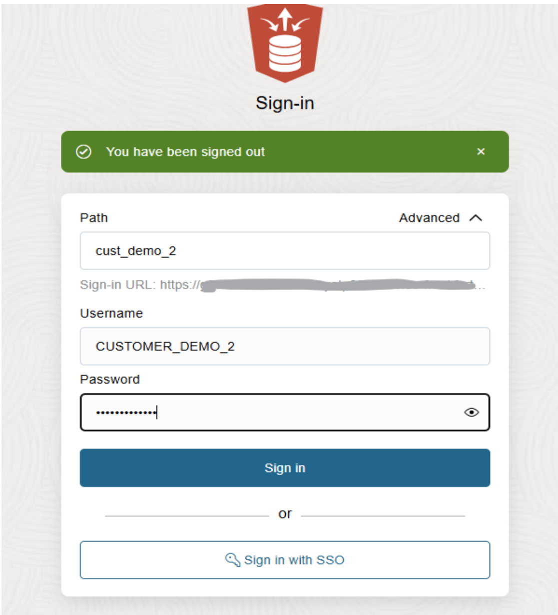
Roles like DWROLE and DB_DEVELOPER_ROLE (which grants the access to Data Studio tools and other developer features) are considered add-ons.

Note: for data loading from external sources activities (CSV, XLS..), at least DWROLE is recommended.

Sign-In to Oracle Database Actions

You can sign-in to Database Actions by connecting to a single default database or to multiple databases or by using a schema alias.

- Sign-in to DB Actions on ADB via ORDS Schema alias which is different from schema user name:



Here on ***Advanced → Path***, enter the schema alias ("*cust_demo_2*") defined in the ORDS_ADMIN.ENABLE_SCHEMA PL/SQL API and in the Username/password the database schema user and corresponding password. The ORDS alias configuration for this schema user (REST-enabled) can be found as follows (run as ADMIN and query DBA_ORDS_SCHEMA data dictionary view):

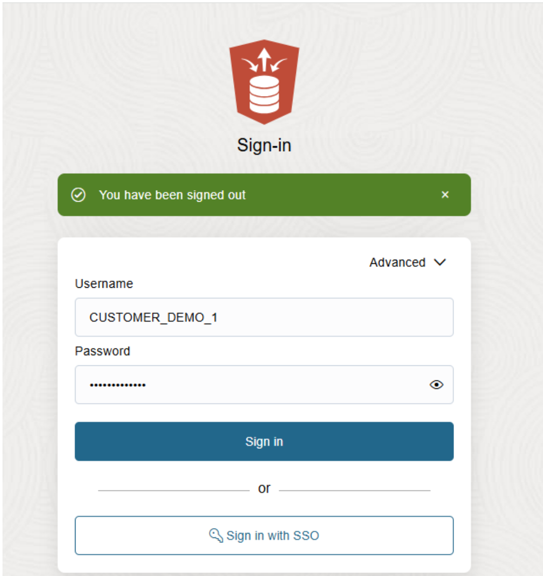
```sql

select parsing_schema, pattern from dba_ords_schemas where parsing_schema = 'CUSTOMER_DEMO_2';

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULT SQL HISTORY TASK MONITOR

All rows fetched: 1 in 0.062 seconds

	PARSING_SCHEMA	PATTERN
1	CUSTOMER_DEMO_2	cust_demo_2

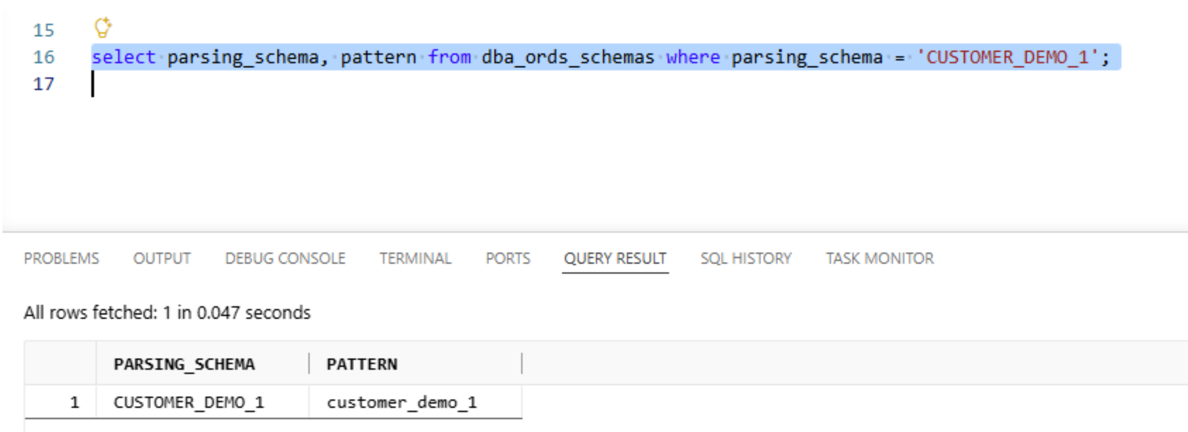
- Sign-in to DB Actions on ADB via ORDS Schema alias with the same name as schema user:



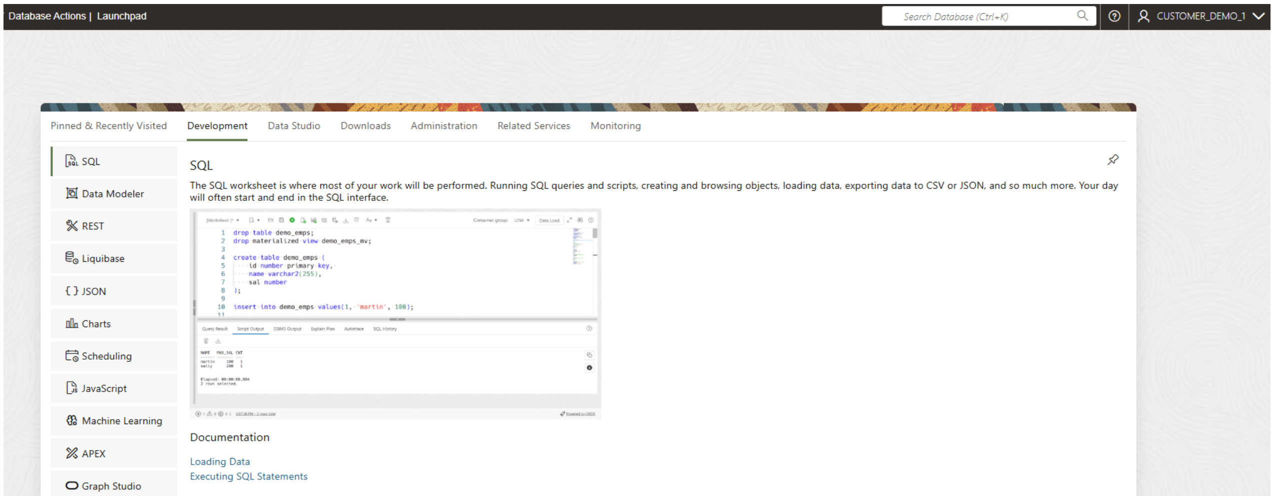
Here, the schema user has been REST-enabled having the alias with the same name as user schema (as ORDS_ADMIN.ENABLE_SCHEMA PL/SQL API shows as follows):

```
ORDS_ADMIN.ENABLE_SCHEMA(
    p_enabled => TRUE,
    p_schema => 'CUSTOMER_DEMO_1',
    p_url_mapping_type => 'BASE_PATH',
    p_url_mapping_pattern => 'customer_demo_1',
    p_auto_rest_auth=> TRUE
);
```

Here, there's no need to enter the schema alias in the Advanced → Path as the schema alias is equal to Database user Schema (not recommended as best practice). If we query the data dictionary view we can notice:



Once you connect to Database Actions (whatever the schema alias scenario) the main DB Actions page is as follows:

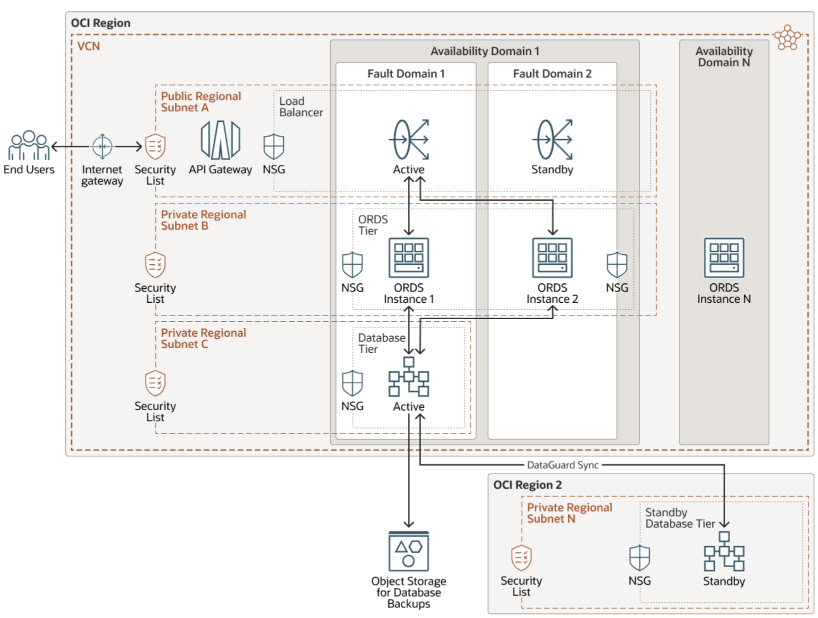


In which the main tools are the SQL Worksheet, Monitoring and REST. All of these options are allowed depending on the roles/privileges assigned, by ADMIN, to current schema user.

Note: as best practice, it is recommended to choose a REST-alias different from the effective user schema name so not to expose, in the URL, the user schema name reference.

Oracle Database Actions: Database Cloud Service/ExaDB/on-prem DB

In a non-Oracle Autonomous Database, ORDS layer is installed separately in a compute node (as Standalone or integrated in a Web layer like Tomcat or WebLogic) and once configured more than one database connection configuration can be created (*db-pool*) towards multiple databases. Here a reference architecture about Oracle REST Data Services deployment configuration in High Availability:



Once configured a [database connection on ORDS](), the first db-pool occurrence is identified as *default*. Any additional one is assigned a proper name (example of a non-default db pool) :

```
Database pool: mydb03tc_mypdb1tc

Setting                               Value                                         Source
-----------------------------------   ---------------------------------------------  -----------
database.api.enabled                  true                                          Global
db.connectionType                     basic                                         Pool
db.hostname                           [redacted]                                    Pool
db.password                           ******                                        Pool Wallet
db.port                               1521                                          Pool
db.servicename                        MYPDB1TC.[redacted].myvcn1.oraclevcn.com      Pool
db.username                           ORDS_PUBLIC_USER                              Pool
feature.sdw                           true                                          Pool
restEnabledSql.active                 true                                          Pool
security.requestValidationFunction    ords_util.authorize_plsql_gateway             Pool
standalone.doc.root                   /etc/ords/config/global/doc_root              Global
standalone.http.port                  8080                                          Global
```

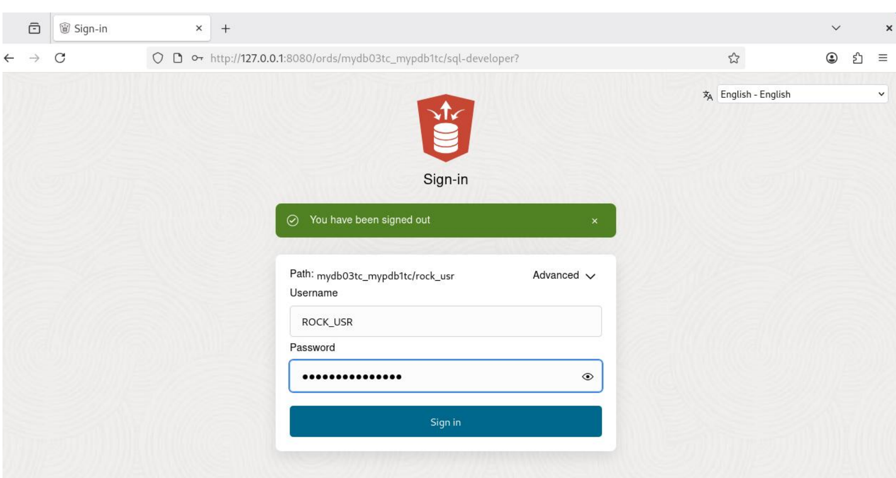
To connect to a Database Pool with Database Actions:

- Verify first that the user you're authenticating with is REST-Enabled

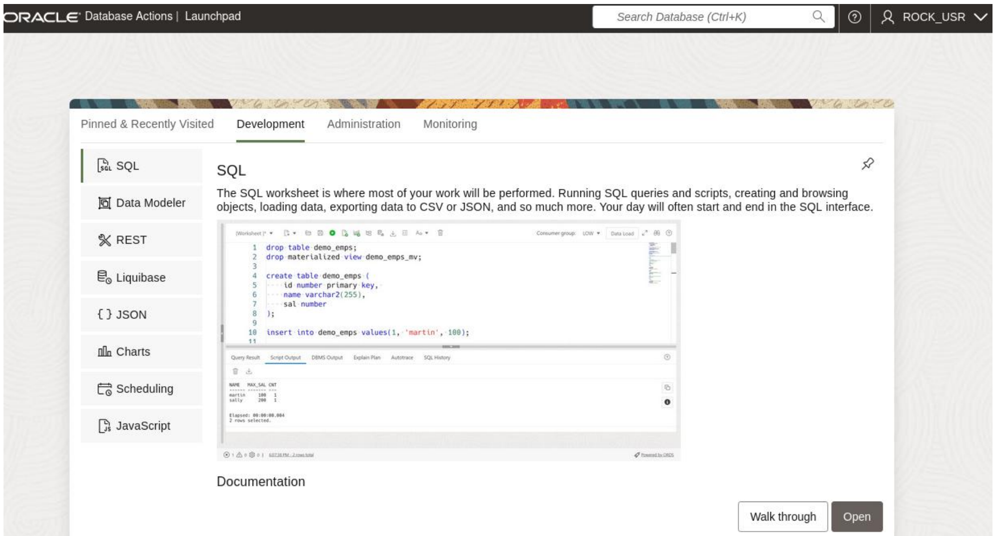
  To Enable a user to connect to Database Actions in a non-ADB database connect with a privileged user (sys as sysdba for example) and run the following:

```
SQL> BEGIN
  2    ords_admin.enable_schema(
  3      p_enabled => TRUE,
  4      p_schema => 'ROCK_USR',
  5      p_url_mapping_type => 'BASE_PATH',
  6      p_url_mapping_pattern => 'rock_usr',
  7      p_auto_rest_auth => TRUE
  8    );
  9    commit;
 10  END;
 11* /
```

- One done the REST-enablement, In the Sign-In page go to ***Advanced → Path*** and in the Path Field enter the database pool name (the non-default one) and schema name in the format: ***<database_pool_name>/<schema_name>*** and then on username and password enter the schema name (in UPPERCASE) and relative password:



In this example, the db-pool configured via ORDS is named mydb03tc_mypdb1tc. Once connected, you can access DB Actions functionalities:



In conclusion, Oracle SQL Developer/Database Actions brings all of your favorite Oracle Database desktop tool's features and experience to your browser. It is already included in Oracle Autonomous AI Database and available with Oracle REST Data Services to non-ADB and on-premise Database.

Coming up soon, as next section to describe the access methods, the Oracle Cloud Infrastructure Database Tools Service.